PolicyGraphAudit-RT

Compliance Audit Report

Heterogeneous GNN Privacy Analysis

Hanu Edu

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Developer	Efficient Way LLC
Genre	BUSINESS
Installs	100+
Audit Date	2026-04-26

Overall Risk Score **0.35** **MEDIUM**

6 Consistent	1 Mismatches	5 Over-disclosure	3 Undeclared

Research prototype · Ongoing research | Policy: https://hanuedu.clubi.mn/terms_and_condition.html

Executive Summary

Hanu Edu declares 3 data type(s) collected and 0 shared in its Play Data Safety label, across 15 data type(s) analyzed. Our heterogeneous GNN audit model identifies **3 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **1 policy-vs-label mismatch(es)**. Overall risk score: **0.35 (MEDIUM)**.

Top discrepancies by risk:

- **installed_apps**: Undeclared Collection (confidence 0.99)
- **crash_logs**: Undeclared Collection (confidence 0.99)
- **advertising_id**: Undeclared Collection (confidence 0.97)

Class definitions:

CONSISTENT: Label, policy, and SDK signals agree.

POLICY_LABEL_MISMATCH: Label discloses data not mentioned in policy.

OVER_DISCLOSURE: Policy mentions data not declared in label.

UNDECLARED_COLLECTION: SDK signals collection undisclosed in label and policy.

Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

Data Type	Class	Conf.	Evidence
installed_apps	**Undeclared Collection**	0.99	label:N policy:N sdk:Y (3, 5, 7 +1)
crash_logs	**Undeclared Collection**	0.99	label:N policy:N sdk:Y (3, 5, 7 +1)
advertising_id	**Undeclared Collection**	0.97	label:N policy:N sdk:Y (3, 5, 7 +1)
name	**Policy Label Mismatch**	1.00	label:Y policy:N sdk:Y (3, 5, 7 +1)
sms_mms	**Over Disclosure**	0.98	label:N policy:Y sdk:N (3, 5, 7 +1)
date_of_birth	**Over Disclosure**	0.98	label:N policy:Y sdk:N (3, 5, 7 +1)
fitness_info	**Over Disclosure**	0.91	label:N policy:Y sdk:N (3, 5, 7 +1)
purchase_history	**Over Disclosure**	0.88	label:N policy:Y sdk:N (3, 5, 7 +1)
other_info	**Over Disclosure**	0.76	label:N policy:Y sdk:N (3, 5, 7 +1)
device_id	**Consistent**	1.00	label:N policy:Y sdk:Y (3, 5, 7 +1)
diagnostics	**Consistent**	1.00	label:N policy:Y sdk:Y (3, 5, 7 +1)
phone_number	**Consistent**	1.00	label:Y policy:Y sdk:N (3, 5, 7 +1)
email_address	**Consistent**	1.00	label:N policy:Y sdk:Y (3, 5, 7 +1)
photos	**Consistent**	1.00	label:Y policy:Y sdk:N (3, 5, 7 +1)
app_interactions	**Consistent**	0.87	label:N policy:Y sdk:N (3, 5, 7 +1)

Evidence Trails

Detailed evidence for UNDECLARED_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

Discrepancy: installed_apps — Undeclared Collection (confidence 0.99)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 3, 5, 7, 16

Recommended action: Add 'installed_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: crash_logs — Undeclared Collection (confidence 0.99)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 3, 5, 7, 16

Recommended action: Add 'crash_logs' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: advertising_id — Undeclared Collection (confidence 0.97)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 3, 5, 7, 16

Recommended action: Add 'advertising_id' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: name — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 3, 5, 7, 16

Recommended action: Update the privacy policy to disclose 'name' collection, aligning it with the Play Data Safety label.

Discrepancy: sms_mms — Over Disclosure (confidence 0.98)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 3, 5, 7, 16

Recommended action: Review whether 'sms_mms' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: date_of_birth — Over Disclosure (confidence 0.98)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 3, 5, 7, 16

Recommended action: Review whether 'date_of_birth' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: fitness_info — Over Disclosure (confidence 0.91)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 3, 5, 7, 16

Recommended action: Review whether 'fitness_info' is genuinely collected. If yes, add to label; if no, remove from policy.

Discrepancy: purchase_history — Over Disclosure (confidence 0.88)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 3, 5, 7, 16

Recommended action: Review whether 'purchase_history' is genuinely collected. If yes, add to label; if no, remove from policy.

Methodology

PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md.

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

Disclaimer